

Provision of Contingency Frequency Services for Virtual Power Plants With Aggregated Models

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Abstract—The growth of variable renewable generations will reduce synchronous inertia, and the demand for managing contingency frequency support is growing. Distributed energy resources (DERs) can provide contingency frequency support via the virtual power plant (VPP). Given that the VPP may cover thousands of DERs with diverse characteristics and response latency, this paper proposes to use equivalent aggregation models for the VPP to participate in contingency reserve services. First, the equivalent aggregation model for the VPP's aggregated frequency response is developed, accurately and concisely restoring the DER aggregation's capability to stop the system frequency from falling. On this basis, the VPP's performance-to-cost map is constructed, which specifies the minimal cost to achieve the same performance as different equivalent aggregation models. In order to encourage DERs to report their response latency accurately, additional latency is penalized based on the resulting cost increase. A sensitivity-based method in the matrix form is developed to estimate the marginal increasing cost due to the increasing latency. Numerical studies are conducted to validate the equivalent aggregation model's effectiveness and illustrate how latency affects response performance and aggregation cost.

Index Terms—Distributed energy resources, contingency frequency support, response latency, virtual power plants, DER aggregation, marginal cost, frequency control.

I. INTRODUCTION

LARGE-SCALE integration of renewable generators into the power grid reduces its synchronous inertia and raises the possibility of supply-demand imbalances. As a result, severe contingencies, as characterized by the rate of change of frequency (RoCoF) and the frequency deviations in absolute value, will occur more often [1]. Such contingencies are traditionally addressed through under-frequency load shedding

(UFLS) [2]. However, UFLS trips all loads within a service region without considering the incurred economic losses.

On the demand side, adjustable distributed energy resources (DERs) are flourishing. DERs are controllable energy resources located close to load centers that can be used to provide value to the grid [3]. DERs consist of battery energy storage, electric vehicles, and controllable loads, such as electric heating systems. They are potential resources that can provide frequency support as an alternative to UFLS. Due to the limited scale of individual DER units, they are collectively formed as a virtual power plant (VPP) to provide considerable frequency support capabilities during contingencies [4].

To engage DERs in contingency frequency support services, the system operator does not only need to know the sizes of the DERs' reserves but also should know the reserves' dynamic response performances following contingencies [3]. In the conventional approach, the VPP, on behalf of the DERs, provides the costs and dynamic parameters (including response latency) of individual DERs to the system operator. Then the system operator evaluates the contribution of each DER [5]. In particular, DERs' response latency should be attached with importance since it directly determines whether the DER reserve can be timely activated when a contingency happens [6]. Given the growing number of DERs, the conventional approach adds tremendous complexity to interactions and dispatching for the system operator. Besides, the conventional approach may generate concerns about DERs' privacy [7].

Consequently, an equivalent approach should be developed to facilitate efficient and privacy-persevering interactions between the VPP and the system operator. The equivalent approach substitutes the sum of raw dynamic models of DERs with a single equivalent aggregation model [8]. In contrast to the conventional approach, the system operator and the VPP simply need to communicate with the equivalent aggregation model to characterize the performance of the entire DER aggregation. The challenge inherent in the equivalent approach is to develop a concise but accurate equivalent aggregation model [9]. For contingency frequency support services, the model should utilize as few parameters as possible to accurately capture the transient dynamics of massive DERs with diverse features and response latency [5].

Beyond that, the VPP should also focus on the monetization of contingency frequency reserves. To support its financial decisions and provide the system operator with the cost information, it should construct the performance-to-cost map, which specifies the minimal cost to achieve various

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frequency response performances [10]. Besides, as the VPP's aggregated model is based on DERs' parameters (especially latency) collected by end-users, the aggregation performance, as well as the cost, will be affected by the submitted latency parameters. The VPP should economically penalize DERs whose actual latency is larger than the reported latency so that DERs are encouraged to report the response latency accurately [11].

The equivalent aggregation model for DERs in the VPP has been extensively studied in energy markets and static system operations. On the equivalence for feasible operating P-Q regions of the VPP, a portion of the works used the atomistic bottom-up approach to aggregate feasible operating P-Q regions of individual DERs [12], [13]; other studies used the direct searching method [14] or robust optimization approach [15] to locate the boundary and vertex of feasible operating P-Q regions of the entire VPP. On the equivalent models of DERs' dynamic response, [16] investigated the fine-grained aggregated model incorporating power electronic devices' control loops. Some research focused on DERs' equivalence for primary frequency control characteristics. Reference [17] sought to find the equivalent power-frequency droop curve at the point of interconnection of the distribution network. Reference [18] proposed an equivalent model for DER aggregation's droop coefficient and inertia constant by minimizing the error of frequency trajectory in the full time domain. Reference [19] suggested a single-machine reduction equivalent model for thermal generators. Reference [20] utilized polynomial fitting to depict generators' primary frequency response characteristics and aggregated these polynomials as the equivalent model. This paper focuses on DERs' equivalent aggregation models, which can reflect DERs' frequency support capability as accurately and simply as possible while considering their heterogeneous response latency.

Concerning aggregation costs, several studies have been devoted to incorporating frequency constraints into the economic dispatch model. Reference [21] optimized the secondary frequency regulation schedule considering constraints on the dynamics of generators. Reference [5] optimized instantaneous and ramping reserves to support frequency contingencies with approximate responses. Reference [22] designed the optimal portfolio of energy storage's frequency response services, each of which has distinguished characteristics. Instead of developing a single optimization program, this paper attempts to build a performance-to-cost map for the VPP. The map specifies the VPP's minimal cost to provide reserves at different frequency support performances, which are further represented by different equivalent aggregation models. Beyond that, this paper also investigates the effects of response latency on aggregation costs. Latency has been studied as a crucial factor in power system stability control [23]. Nevertheless, when a longer latency does not cause oscillation [6] and instability, it still increases the VPP's operation costs. Quantitatively evaluating the economic effects of DERs' additional latency is essential.

Against this background, this paper investigates using the equivalent approach to help the VPP interact with the system operator in contingency frequency reserve

services. Specifically, this paper makes the following contributions:

- 1) Develop a holistic equivalent aggregation model to represent the collective responses of DERs in the VPP, and formulate an optimization-based technique to obtain the model's parameters. The aggregation model takes into account heterogeneity in response latency among different DERs, and can accurately and efficiently represent the DER aggregation's transient frequency support capability.
- 2) Construct the performance-to-cost map for the VPP based on the proposed equivalent aggregation model. The map serves as a crucial tool for the system operator to know the aggregated reserve's cost and performance simultaneously. Besides, the map is built via solving lightweight optimization programs parallelly and can be updated online.
- 3) Provide an analytic method for evaluating the marginal increasing cost of DERs' increasing response latency. The marginal cost of latency serves as a crucial tool to punish DERs for deviations from their reported latency. The evaluation method is in matrix form, allowing for rapid computation and eliminating the need to solve optimization problems repeatedly.

The remaining sections are organized as follows: Section II describes the problem of aggregating a large number of DERs by the VPP to participate in contingency frequency reserve services. Section III presents the proposed equivalent aggregation model for DERs with heterogeneous response latency; formulates the optimization problem for the VPP to determine the parameters of the equivalent models. Section IV builds the cost-to-performance map based on the aggregation cost minimization problem, and provides the method to compute the increasing marginal cost of latency. Section V conducts case studies to verify the effectiveness of the proposed method and explicitly analyzes the effects of latency. Section VI draws conclusions and gives future research directions.

II. PROBLEM FORMULATION

There are three main actors in the system: the system operator, the VPP, and DERs, the interactions of which are shown in Fig. 1. This paper considers the configuration of the VPP as a commercial VPP, and DER units are distributed throughout different buses in the distribution system. The system operator maintains a stable operation of the whole power system, preventing major supply-demand imbalances for system stability (such as the loss of a generating unit). When the supply drops far below the demand unexpectedly, the system frequency declines, and a contingency occurs. To prevent the contingency, the system operator arranges the real-time contingency frequency reserve market (usually cleared every 15 minutes) to acquire reserves from different stakeholders, including the VPP. Contingency frequency control is different from primary frequency control. The primary frequency control is intended for normal operating status, which passively acts according to the droop-based

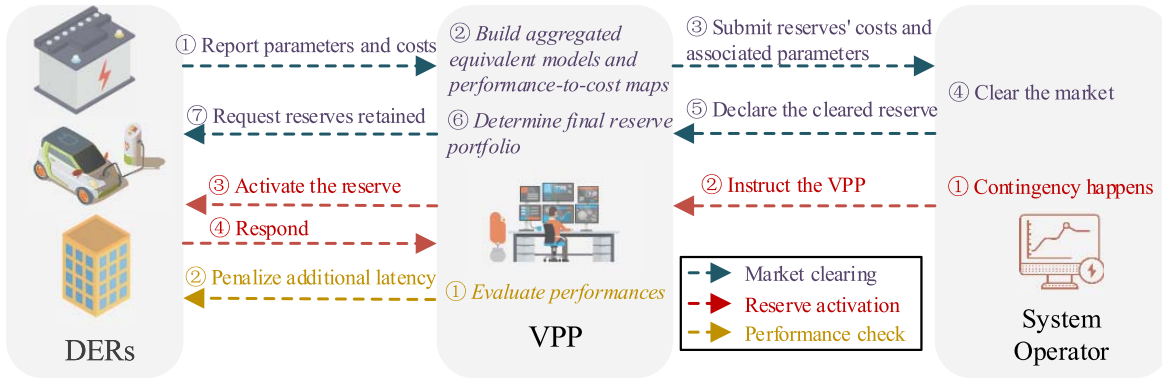


Fig. 1. The market clearing, reserve activation, and performance check phases in contingency frequency services. Different phases are marked in distinct colors. The processes in *Italics* are the scope of this paper.

and inertia-based characteristics to attenuate frequency deviations locally. The contingency control is only activated after a severe load-generation imbalance or fault; it does not provide inertia directly, but actively changes the set-points of different resources to stop the frequency from falling. The contingency frequency reserve is adopted in real power markets, such as Australia's contingency frequency control ancillary services (FCAS) [24].

Each DER can be either storage-type or load-type. Storage-type DERs are connected to the power grid through inverters and can adjust their output continuously. Load-type DERs are connected to the power grid through smart switches and are controlled between on and off states. To participate in contingency frequency services, both types of DERs should meet the following requirements:

- They should maintain a stable base power set-point during one market period (15 minutes). If the base power consumption is volatile and random, the VPP can not know whether the DER truly responds to the instructions to fulfill the service because its power keeps changing.
- They must be turned off immediately after receiving the VPP's instructions. It guarantees that the DERs can provide a fast response when a contingency happens.
- They can sustain being turned off before contingency events clear (for about 2-5 minutes) and they are turned on again.

Typical qualified storage-type DERs include residential energy storage, backup uninterruptible power supplies (UPSs), and electric vehicle batteries. Typical qualified load-type DERs include lighting devices, refrigerators, and air-conditioners in residential houses and buildings.

DERs participate in contingency frequency services through the VPP. The complete process is exhibited in Fig. 1. DERs first report their reserve costs and necessary technical parameters to the VPP. Then, the VPP submits the reserve costs and associated parameters to the system operator on their behalf. After the upper market is cleared, the VPP determines its final contingency frequency reserve portfolio according to the cleared result. The VPP requests the DERs in the selected portfolio to maintain the reserve during the subsequent market cycle. For DERs successfully selected in the final reserve portfolio, it is their duty to guarantee their response performance

is fulfilled. Otherwise, the VPP will suspend the payments to these DERs that fail to activate enough reserves as agreed. When the system operator's control center detects a contingency through its monitoring system, it sends signals to instruct the VPP to activate the reserve.

In a naive approach, the VPP submits all the costs and dynamic parameters of individual DERs to the system operator. The naive approach adds immense complexity for the system operator to dispatch and control reserves, given that the VPP may cover thousands of DERs with diverse characteristics. Contrary to the naive approach, building an equivalent model capturing the collective effects of DERs increases system scalability. The equivalent model captures the collective frequency support performance of DERs. In this way, the VPP only needs to submit the aggregation model, which is more straightforward than the naive approach.

Response latency is a crucial factor that affects the DER aggregation's frequency support capability. The response latency comprises DERs' communication latency and activated latency. DERs learn their response latency through technical tests and report it to the VPP. With a longer latency, DERs cannot deliver the full reserve power and cannot even be activated. Regarding the system frequency, a longer response latency will diminish the DER aggregation's capability to stop the system frequency from falling. In terms of aggregation costs, a longer response latency also increases aggregation costs. The VPP has to activate more reserves to meet the transient performance requirements. The VPP needs to assess the effect of increased latency on the aggregation cost. Then the VPP scores and punishes the additional latency based on the marginally increasing cost to ensure the DERs submit their response latency truthfully.

Notations: Bold *Italics* $\mathbf{x}$ denote vectors. Bold verticals $\mathbf{X}$ denote matrices. ∇_x is the gradient operator with respect to variable x . ∇_{xy}^2 denotes the mixed second-order partial derivatives with respect to variables x and y . ∇_x^2 denotes the second-order partial derivatives with respect to the variable x . $\mathbb{E}_x[\cdot]$ denotes the math expectation with respect to random variable x . $\text{diag}(\mathbf{x})$ denotes the diagonal matrix of the diagonal vector $\mathbf{x}$. $\otimes$ denotes the Kronecker product. s is the complex parameter in Laplace transforms. $\omega(s)$ denotes ω in Laplace domain, and $\omega(t)$ denotes ω in the time domain. $\dot{\omega}(t)$ denotes

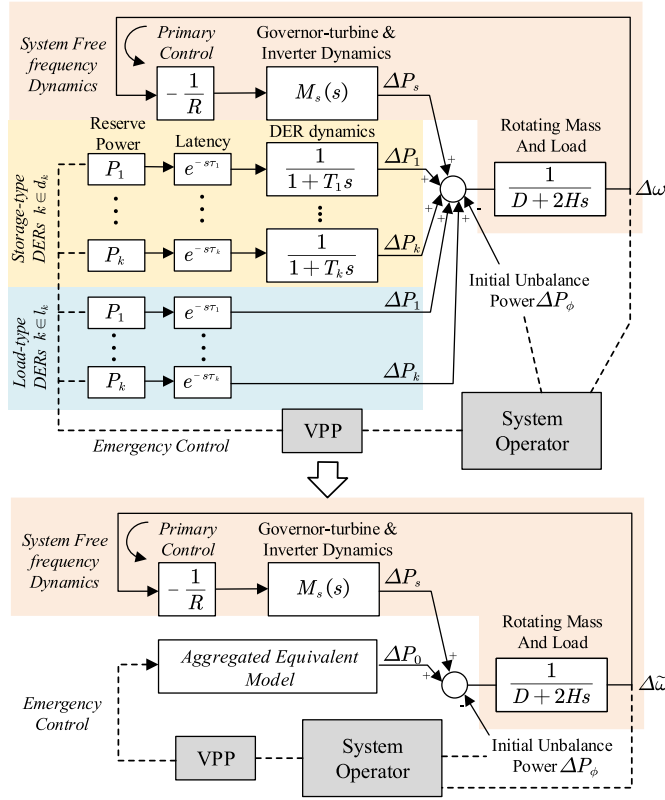


Fig. 2. System architecture and the system frequency response model. The equivalent aggregation model replaces the sum of dynamic models for transient performance analysis. (The blocks in the light-orange area are the POI frequency response model.)

the derivative of ω with respect to time t . $u[t]$ is the step function. $[x_1 : \delta x : x_2]$ denotes the regularly-spaced vector starting from x_1 , ending by x_2 , and using δx as the increment between elements. $\mathbf{x} \times \mathbf{y}$ means the Cartesian product between the vector $\mathbf{x}$ and $\mathbf{y}$.

III. THE EQUIVALENT AGGREGATION MODEL

A. Frequency Response Model

The system frequency response model is shown in Fig. 2. The components are explained as follows:

1) *DERs*: We define DERs as $k = 1, 2, \dots, K$ and let d_k and l_k as the set of storage-type DERs and load-type DERs, respectively. The reported parameters of storage-type DERs are: the time constant T_k , the response latency τ_k , and costs with respect to its reserve capacity $c_k(P_k)$. The first-order model is used for modeling DERs' active power response [18], [25]. The time constant T_k represents DER k 's response speed to change power set-points. A larger T_k indicates it will take DER k a longer time to reach the new set-point. The collected parameters of load-type DERs are: the response latency τ_k , and costs with respect to its reserve capacity $c_k(P_k)$.

2) *POI Frequency Response Model*: The system operator gives the frequency response model at the point of interconnection (POI) to the VPP, shown in Fig. 2. The model is widely adopted to represent the free response of frequency

dynamics $\Delta\omega(s)$ of the whole system following a load-generation imbalance. With the model, the VPP can evaluate how its reserve will influence the system frequency seen by the system operator. In this way, the VPP and the system operator can reach a consensus on how much contribution the VPP can make to system frequency dynamic enhancement. Related parameters and models are obtained online from the results of the dynamic state estimation conducted by the system operator.

The POI frequency dynamic response model and its parameters include: the governor-turbine and inverter dynamics $M_s(s)$; the rotating mass and load effect $1/(D + 2Hs)$ where D is the load damping ratio, and H is the inertia constant; and the primary control effect $-1/R$ with a droop characteristic of R . The model typically consists of less than 10 parameters. It is simple and will not leak confidential system information.

$M_s(s)$ reflects the collective dynamics of the system's thermal generators and renewable generations. $M_s(s)$ has been modeled as zero (so the system frequency dynamic is a simplified swing equation) [5], [22], or the first-order generator model [26], or the second-order generator model [27], and so on. This paper does not restrict the specific expression of $M_s(s)$ such that the models above can all be accommodated. The system operator should determine its form. Besides, the system operator should also specify the prediction about contingency imbalance ΔP_ϕ 's distribution, and the candidate set $\mathcal{P}$ of equivalent aggregation models, which will be discussed later.

3) *Overall Frequency Response*: The transfer function of frequency deviations $\Delta\omega(s)$ concerning power injection following a contingency change $\Delta P(s)$ can be obtained from Fig. 2, which is represented as a sum of partial fractions:

$$\frac{\Delta\omega(s)}{\Delta P(s)} = \frac{(D + 2Hs)^{-1}}{1 + (D + 2Hs)^{-1}R^{-1}M_s(s)} = \sum_i \frac{A_i}{s - p_i} \quad (1)$$

where p_i is the pole which may be a real or complex conjugate pair; A_i is the numerator constant corresponding to the pole i obtained from the residue method. Generally, the system is causal and there are no repeating poles, so Eq. (1) holds [28]. The total power injection change $\Delta P(s)$ is composed of the initial power imbalance $-\Delta P_\phi(s)$ and DER k 's power set-point adjustment $\Delta P_k(s)$ as:

$$\Delta P(s) = -\Delta P_\phi(s) + \sum_k \Delta P_k(s) \quad (2)$$

where the feedback primary frequency control effect ΔP_s is already contained in Eq. (1). Combining Eqs. (1)-(2), the total response of the system frequency dynamic trajectory can be decomposed as the contribution of different DERs:

$$\Delta\omega(s) = -\Delta\omega_\phi(s) + \sum_k \Delta\omega_k(s) \quad (3)$$

As shown in Fig. 2, the activated power set-point adjustments following the VPP's instructions are expressed as:

$$\Delta P_k(s) = \begin{cases} \frac{P_k}{s} \frac{1}{1 + sT_k} e^{-s\tau_k}, & \forall k \in d_k \\ \frac{P_k}{s} e^{-s\tau_k}, & \forall k \in l_k \end{cases} \quad (4)$$

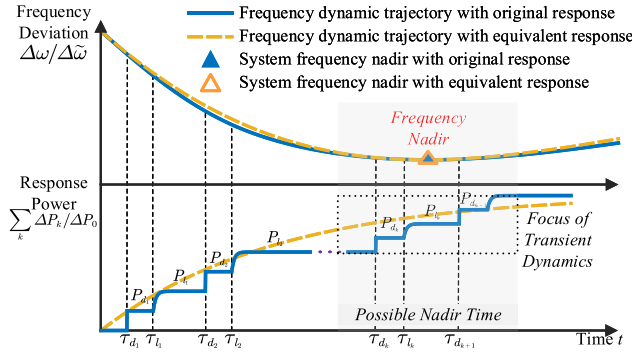


Fig. 3. The equivalent aggregation model minimizes frequency errors at possible nadir time, and it should be close to the original response power at possible nadir time.

An example of the DER aggregation power response curve is shown in Fig. 3.

Combining Eq. (1) and Eq. (4), the contribution of DER k to frequency changes $\Delta\omega_k(s)$ is expressed as:

$$\Delta\omega_k(s) = \begin{cases} \frac{P_k}{s} \frac{e^{-s\tau_k}}{1+sT_k} \sum_i \frac{A_i}{s-p_i}, & \forall k \in d_k \\ \frac{P_k}{s} e^{-s\tau_k} \sum_i \frac{A_i}{s-p_i}, & \forall k \in l_k \end{cases} \quad (5)$$

After factorizing the denominator of Eq. (5) and performing the inverse Laplace transform, the time domain expression $\Delta\omega_k(t)$ is shown in Eq. (6), where it is assumed that $p_i T_k \neq -1$. Similarly, the derivative of $\Delta\omega_k(t)$ with respect to time t is shown in Eq. (7). Eq. (6) and Eq. (7) are shown in the bottom of the page.

The initial load imbalance $\Delta P_\phi(s)$ during contingencies is often treated as a step function $\Delta P_\phi(s) = P_\phi e^{-s\tau_\phi}/s$. Consequently, $\Delta P_\phi(s)$ can be expressed in the same form as Eq. (4) for $k \in l_k$ with $\tau_\phi = 0$. The expressions of $\Delta\omega_\phi(t)$ and $\Delta\dot{\omega}_\phi(t)$ are obtained from Eq. (6) and Eq. (7), respectively.

B. Transient Performance Equivalent Model

The equivalent model $\Delta P_0(s)$ is selected in the candidate set of $\mathcal{P}$, which is offered by the system operator. If $\Delta P_0(s)$ in $\mathcal{P}$ is equivalent with $\sum_k \Delta P_k(s)$, it means the system operator thinks $\sum_k \Delta P_k(s)$ and $\Delta P_0(s)$ in $\mathcal{P}$ achieve the same transient performance in contingency frequency regulations services. In that way, the system operator simply deals with $\Delta P_0(s)$ instead of $\sum_k \Delta P_k(s)$.

Suppose the VPP provides aggregated equivalent response $\Delta P_0(s)$ in place of the naive response $\sum_k \Delta P_k(s)$, the system frequency dynamic is supposed to evolve as $\Delta\tilde{\omega}(t)$ compared

with $\Delta\omega(t)$ defined in Eq. (3):

$$\Delta\tilde{\omega}(t) = -\Delta\omega_\phi(t) + \Delta\omega_0(t) \quad (8)$$

By comparing the system frequency dynamics between Eq. (8) and Eq. (3), the definition of the equivalent aggregation model for DERs is presented as follows:

Definition 1 (Transient Performance Equivalence): The transient performance equivalence $\Delta P_0(s)$ for DERs' aggregation $\sum_k \Delta P_k(s)$ is the response in the candidates $\mathcal{P}$ that achieves the same frequency nadir time while achieving the closest frequency drop with $\sum_k \Delta P_k(s)$. Namely, $\Delta P_0(s)$ is the solution to the following optimization problem:

$$\min_{\Delta P_0(s) \in \mathcal{P}} \mathbb{E}_{P_\phi} \left[\left(\Delta\tilde{\omega}(t_\phi^*) - \Delta\omega(t_\phi^*) \right)^2 \right] \quad (9)$$

$$\text{s.t. } \Delta\dot{\tilde{\omega}}(t_\phi^*) = 0, \Delta\dot{\omega}(t_\phi^*) = 0, \forall \phi \quad (10)$$

where t_ϕ^* is the frequency nadir time when the initial imbalance is P_ϕ . Because the initial imbalance is unknown, the expectation is used with the predicted distribution of P_ϕ offered by the system operator.

Various functions can be used as the prototype for the candidate set $\mathcal{P}$, where one should trade-off its accuracy and complexity. A qualified candidate set $\mathcal{P}$ should provide enough freedom to guarantee that the equivalent model can accurately restore the aggregation's response capability; it also should be simple for communication and computation. This paper uses the set of first-order process as the prototype function for the candidate set, namely $\mathcal{P} = \{P_0/(s(1+T_0s)) | P_0 \geq 0, T_0 \geq 0\}$. Through experiments, the selected candidate set can provide satisfying accuracy with only two parameters T_0 and P_0 , compared with at most $3 \times K$ parameters (K groups of T_k, τ_k, P_k) of all the DERs. Because the equivalent aggregation model and the storage-type DER's dynamic model are both first-order processes, $\Delta\omega_0(t)$ and $\Delta\dot{\omega}_0(t)$ are in the same form of Eqs. (6)-(7) for $k \in d_k$, respectively. In that way, the optimization problem Eqs. (9)-(10) is to find the equivalent parameters T_0 and P_0 .

The optimization problem Eqs. (9)-(10) is hard to solve due to its nonlinear constraints. The definition of equivalence Def 1 is relaxed by the penalized function method to simplify the solution process:

Definition 2 (λ -Penalized Equivalence): The λ -penalized transient performance equivalence $\Delta P_0(s)$ for the DER aggregation $\sum_k \Delta P_k(s)$ is the response in the candidate set $\mathcal{P}$ that solves the following penalized optimization problem with penalized coefficient λ :

$$\min_{\Delta P_0(s) \in \mathcal{P}} \mathcal{F} = \mathbb{E}_{P_\phi} \left[(\Delta\tilde{\omega} - \Delta\omega)^2 \right] + \frac{\lambda}{2} \|\Delta\dot{\tilde{\omega}}\|^2 \quad (11)$$

$$\Delta\omega_k(t) = \begin{cases} P_k \sum_i \left[\frac{A_i}{p_i} (e^{p_i(t-\tau_k)} - 1) - \frac{A_i}{p_i + T_k^{-1}} (e^{p_i(t-\tau_k)} - e^{-T_k^{-1}(t-\tau_k)}) \right] u[t - \tau_k], & \forall k \in d_k \\ P_k \sum_i \left[\frac{A_i}{p_i} (e^{p_i(t-\tau_k)} - 1) \right] u[t - \tau_k], & \forall k \in l_k \end{cases} \quad (6)$$

$$\Delta\dot{\omega}_k(t) = \begin{cases} P_k \sum_i \frac{A_i}{p_i T_k + 1} [e^{p_i(t-\tau_k)} - e^{-T_k^{-1}(t-\tau_k)}] u[t - \tau_k], & \forall k \in d_k \\ P_k \sum_i [A_i e^{p_i(t-\tau_k)}] u[t - \tau_k], & \forall k \in l_k \end{cases} \quad (7)$$

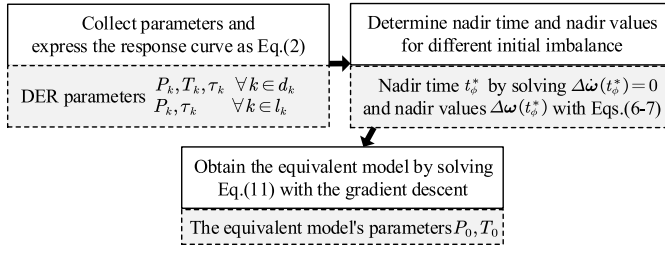


Fig. 4. The workflow diagram to establish the equivalent aggregation model for DERs. The grey boxes are the results in each step.

Since t_ϕ^* could be solved numerically with equation $\Delta\dot{\omega}(t_\phi^*) = 0$ directly, so t_ϕ^* and $\Delta\omega(t_\phi^*)$ are treated as known parameters and only the constraint $\Delta\dot{\omega}(t_\phi^*) = 0$ gets penalized.

From Def 1 to Def 2, the penalty function method is used. Through this conversion, Eqs. (9)-(10) are transformed into an unconstrained optimization problem shown as Eq. (11), which can be further solved through numerical methods such as gradient descents. For the candidate set $\mathcal{P} = \{P_0/(s(1 + T_0s))\}$, the optimization problem Eq. (11) can be efficiently solved through the gradient descent algorithm with respect to variables P_0 and T_0 . The expressions of the gradients are shown in the Appendix. Def 2 does not strictly require the nadir time must be the same but requires that the frequency nadir time of the equivalent dynamic response is close to that of the original dynamic response. The larger λ is, the closer the nadir time will be. By increasing the penalty factor $\lambda \rightarrow \infty$, the solutions to the penalized optimization problem will approach the feasible region in Def 1. Ideally, as $\lambda \rightarrow \infty$, the solution point of the penalized problem will converge to a solution of the constrained problem [29].

In summary, the workflow to establish the equivalent aggregation model for DERs is shown in Fig. 4, which concretely contains the following steps:

- The VPP first collects related parameters of DERs to constitute the original response power curve as Eq. (2). The parameters collected are DERs' response power P_k , response latency τ_k , and storage-type DERs' time constants T_k .
- With the original response power curve, the VPP determines frequency nadir time t_ϕ^* and nadir values $\Delta\omega(t_\phi^*)$ via Eqs. (6)-(7) under different initial imbalance power P_ϕ . t_ϕ^* and $\Delta\omega(t_\phi^*)$ are key indicators for the transient response performance of the DER aggregation.
- The VPP takes the values of t_ϕ^* and $\Delta\omega(t_\phi^*)$ as parameters of the model Eq. (11) and solves Eq. (11) with the gradient descent method. Finally, the equivalent aggregation model's parameters P_0, T_0 are obtained.

C. Comparisons and Advantages

Table I compares the proposed equivalent aggregation model to models from previous studies, where * denotes the equivalent aggregation model presented in this paper. Different models are categorized based on the adopted equivalence methodologies (the last column). The equivalence methodology refers to the method to obtain the equivalent model's

TABLE I
SUMMARY OF DIFFERENT EQUIVALENT AGGREGATION MODELS

	Latency	Equivalence Model	Equivalence Methodology
[16]	No	Control loop	Atomistic
[17]	No	$P - \omega$ droop curve	Atomistic
[19]	No	State-space model	Atomistic
[20]	No	Polynomial function	Atomistic
[5]	Yes	Ramping/step function	Atomistic
[22]	Yes	Ramping function	Atomistic
[18]	No	State-space model	Holistic
*	Yes	Transfer function	Holistic

parameters. References [16], [17], and [19] directly summed or averaged DER units' parameters to obtain the equivalent aggregation model. For example, the aggregation power is the sum of that of DER units, and the aggregated time constant is the mean. References [5], [20], and [22] utilized curve fitting and then summed the curves. Reference [18] and this paper employed an optimization-based method to find dynamic models that minimized different performance metrics. In Table I, it is shown that most studies, as well as this work, utilize dynamic models (including transfer functions or state-space models) as the final equivalent aggregation model.

The first weakness of previous equivalent methods is the ignorance of the response latency. Most equivalent models are proposed for local primary frequency regulations. These models ignore the response latency that must be accounted for in contingency frequency services. The *directly sum/average* method performs quite well without DERs' heterogeneous response latency according to the results in [16], [17], [19]. Nevertheless, it fails when DERs' heterogeneous response latency is included, which will be shown in the case study.

The second weakness is the accumulation of error induced by the bottom-to-up atomistic approach. Several studies employed an atomistic approach to building the equivalent model. They first specified a prototype model for each DER and then summed the prototype models as the aggregation model. The inaccuracy of the prototype model will be accumulated in the aggregation model. Nevertheless, this paper employs a holistic approach. The proposed method directly finds the equivalent model for the aggregation of DERs to avoid accumulated errors. Besides, the atomistic approach using the ramping function as the prototype model requires 2K-3K parameters to represent the aggregation model, which increases the parameter complexity.

The third problem is the out-of-focus to accurately capture the DER aggregation's transient performance. Our proposed method concentrates on finding the most accurate equivalent aggregation model of the transient response performance. An illustration is shown Fig. 3. The equivalent response power should be as close to the original response power as possible during possible nadir time to reduce the nadir time difference $|t_\phi^* - t_\phi^*|$ and the nadir value difference $|\Delta\tilde{\omega}(t_\phi^*) - \Delta\omega(t_\phi^*)|$ with the original frequency trajectory. Meanwhile, some methods [18] may fail because it minimized system frequency differences $|\Delta\tilde{\omega} - \Delta\omega|$ in the entire time domain (from $t = 0$

to $t = \infty$), which overemphasized DERs' response after frequency nadir time till $t = \infty$.

IV. PERFORMANCE-TO-COST MAP AND LATENCY COST

This section constructs the performance-to-cost map and computes the effect of increasing latency on the aggregation cost. The aggregation cost minimization problem is first established as the basis of the map. It offers the minimal cost for the VPP to deliver the contingency reserve whose performance is the same as an equivalent aggregation model with parameters P_0 and T_0 . Then, the complete process to construct the performance-to-cost map is illustrated. Besides, the marginal increasing cost due to increasing latency is computed.

A. Performance-to-Cost Map

1) Cost Minimization With Equivalent Aggregation Models:

The aggregation cost minimization problem needs to be formulated to seek for DERs' portfolio that achieves the lowest cost to ensure a specific aggregated response performance. The complete problem is depicted as follows:

$$\min_{P_k} C = \sum_{k \in l_k} c_k(P_k) + \sum_{k \in d_k} c_k(P_k) \quad (12)$$

$$\text{s.t. } P_0, T_0 = \arg \min_{\Delta P_0(s) \in \mathcal{P}} \mathcal{F} \quad (13)$$

$$P_l = x^T P_{l(\max)}, P_d = y^T P_{d(\max)} \quad (14)$$

$$x \in \{0, 1\}, y \in [0, 1] \quad (15)$$

The objective Eq. (12) is to minimize the total cost $c(P_k)$ of the response reserve P_k of DERs. The constraints Eqs. (13)-(15) are explained as follows:

Eq. (13) is the performance requirement constraint. In the workflow Fig. 4 to obtain the equivalent aggregation model, the reserve portfolio P_k is known as parameters, and the equivalent model's parameters P_0 and T_0 are undetermined. P_0 and T_0 are obtained by solving Eq. (11). In contrast, the performance constraint Eq. (13) uses Def 2 in a reversed way. P_0 and T_0 are given parameters representing the reserve's performance requirements, while the reserve portfolio P_k is undetermined. Eq. (13) first designates the equivalent model's parameters as P_0 and T_0 , explicitly imposing a restriction on the performance of the undetermined DER portfolio P_k . Namely, the combinations of DERs P_k must provide the same transient performance as the equivalent aggregation model with the parameters P_0 and T_0 . Because Eq. (13) is inherently an optimization program, the entire problem becomes a bi-level optimization problem.

Eqs. (14)-(15) are DERs' adjustment constraints. x is the binary variable indicating whether a load-type DER is selected; y is the continuous variable indicating the selected amount of a storage-type DER's reserve. $P_{\max}$ is the rating power for load-type DERs, and storage-type DERs' maximum reserve submitted by DERs' owners. The constraints for the energy storage's energy level are not considered because they should already be accounted for when DERs' owners submit their available reserve capacity to the VPP. Besides, the contingency frequency support is less frequently triggered than conventional frequency regulation services and will not last long after

TABLE II
COMPARISONS AMONG DIFFERENT EQUIVALENCE DEFINITIONS

Def	Nadir Value Difference	Nadir Time Difference	Advantage	Disadvantage
1	Minimum	Zero	Closest equivalence	Constrained problem and hard to solve
2	Weighted minimum	Weighted minimum	Unconstrained problem, can be solved numerically	Still is a nonlinear optimization problem
3	Within a tolerance	Within a tolerance	Linear, can be embedded in optimization problems	Needs to tune the relaxed tolerance level

being activated. Therefore, the energy level of energy storage will not be deeply affected.

The cost function of load-type DER l_k is defined as $c_k(P_{l_k}) = \varrho_{l_k} P_{l_k}$. It indicates if load-type DER l_k is selected as the final reserve, it will respond with power P_{l_k} and incur cost ϱ_{l_k} . For storage-type DER d_k , its cost function is defined in quadratic functions as P_{d_k} as $c_k(P_{d_k}) = \rho_{d_k} P_{d_k}^2 + \varrho_{d_k} P_{d_k}$.

2) *Solution Method*: The problem Eqs. (12)-(15) is a bi-level programming problem with binary variables. To construct the performance-to-cost map, the VPP needs to solve Eqs. (12)-(15) for different performance requirements, namely for different pairs of T_0 and P_0 . It requires the optimization problem to be lightweight and efficiently solved so that the VPP can construct the map online and submit it to the system operator. To simplify the solution process, this paper further proposes the $(\varepsilon_1, \varepsilon_2)$ -relaxed equivalence relaxed from Def 2:

Definition 3 [$(\varepsilon_1, \varepsilon_2)$ -Relaxed Equivalence]: $\Delta P_0(s)$ is said to be in $(\varepsilon_1, \varepsilon_2)$ -relaxed equivalent with $\sum_k \Delta P_k(s)$, if their transient performances are similar within tolerance ε_1 and ε_2 :

$$\left| \Delta \omega(\tilde{t}_\phi^*) - \Delta \tilde{\omega}(\tilde{t}_\phi^*) \right| \leq \varepsilon_1, \left| \Delta \dot{\omega}(\tilde{t}_\phi^*) \right| \leq \varepsilon_2, \forall \phi \quad (16)$$

where $\tilde{t}_\phi^*$ is the nadir time with the equivalent response power and satisfies $\Delta \tilde{\omega}(\tilde{t}_\phi^*) = 0$; ε_1 is a small tolerance constant to ensure the frequency value at time $\tilde{t}_\phi^*$ close; ε_2 is a small tolerance constant to ensure the frequency nadir time is close.

By replacing the constraint Eq. (13) by Eq. (16), the bi-level problem is converted to a single-level problem. Besides, after taking parameters into Eq. (6) and Eq. (7), $\Delta \omega_k(\tilde{t}_\phi^*)$ and $\Delta \dot{\omega}_k(\tilde{t}_\phi^*)$ are just a linear combination of P_k :

$$\Delta \omega(\tilde{t}_\phi^*) = a_l^T P_l + a_d^T P_d + a_\phi \quad (17)$$

$$\Delta \dot{\omega}(\tilde{t}_\phi^*) = b_l^T P_l + b_d^T P_d + b_\phi \quad (18)$$

where a_l , a_d , b_l , b_d are linear coefficients obtained by taking parameters into Eq. (6) and Eq. (7), respectively. a_ϕ , b_ϕ are constants obtained by taking parameters (including the value of P_ϕ) into Eq. (6) and Eq. (7), respectively. After relaxation, the optimization model is converted to a mixed-integer quadratic programming (MIQP) problem. It can be solved efficiently by using commercial solvers.

Def 3 converts the equivalent model from a nonlinear optimization problem to a set of linear inequalities, which can be easily embedded in another optimization program. A comparison among three different definitions of equivalence is shown in Table II. From Def 1 to Def 3, the key changes are the requirements on the differences between the equivalent model's nadir value/ time and the original nadir value/ time.

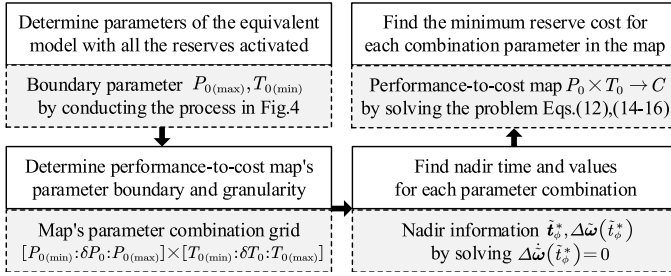


Fig. 5. The workflow diagram to build the cost-to-performance map. The grey boxes are the results in each step.

The complexity of equivalence is reduced at the cost of a more slack requirement on accuracy of the equivalence performance. It is expected that a tighter tolerance can make the frequency nadir time close enough ($\varepsilon_2 \rightarrow 0$), and the frequency nadir value is close enough ($\varepsilon_1 \rightarrow 0$), Def 3 will approach Def 1.

3) *Complete Process of Map Construction*: In summary, the workflow to build the performance-to-cost map $P_0 \times T_0 \rightarrow C$ is shown in Fig. 5. The steps are explained as follows:

- The VPP first assumes that all the reserves will be activated when the contingency happens. Under this extreme circumstance, the VPP finds the parameters $P_0 = P_{0(\max)}$, $T_0 = T_{0(\min)}$ of the equivalent model according to the workflow shown in Fig. 5. ($P_{0(\max)}$, $T_{0(\min)}$) is the highest-performance boundary point in the parameter space. Namely, the VPP cannot provide a contingency frequency reserve larger than $P_{0(\max)}$ and faster than $T_{0(\min)}$ simultaneously.
- The VPP selects another boundary point ($P_{0(\min)}$, $T_{0(\max)}$) that serves as the lowest-performance reserve it attempts to provide. Then The VPP selects the parameter space granularity δP_0 and δT_0 . Finally, the performance parameter grid $[P_{0(\min)} : \delta P_0 : P_{0(\max)}] \times [T_{0(\min)} : \delta T_0 : T_{0(\max)}]$ is constituted. For example, the VPP can select $P_{0(\min)} = 0.5P_{0(\max)}$, $T_{0(\max)} = 5T_{0(\min)}$, $\delta P_0 = 0.1P_{0(\max)}$, and $\delta T_0 = 1T_{0(\max)}$. Then the parameter grid is constituted by $6 \times 5 = 30$ pairs of (P_0 , T_0).
- For each pair of (P_0 , T_0), the equivalent aggregation response power is $\Delta P_0(s) = P_0/(1+T_0s)$. The frequency nadir time $\tilde{t}_\phi^*$ and nadir values $\Delta\tilde{\omega}(\tilde{t}_\phi^*)$ are computed with the equivalent aggregation response power. $\tilde{t}_\phi^*$ and $\Delta\tilde{\omega}(\tilde{t}_\phi^*)$ are key requirements for the transient response performance of the DER portfolio P_k .
- For each pair of (P_0 , T_0), the VPP takes the values of $\tilde{t}_\phi^*$ and $\Delta\tilde{\omega}(\tilde{t}_\phi^*)$ as parameters of the aggregation cost minimization problem solves the problem Eqs. (12), (14)-(16). When the aggregation cost for each pair (P_0 , T_0) is computed, the performance-to-cost map is built.

Every time DERs' owners submit and renew cost and performance parameters (every 15 minutes), the VPP should update the map. To perform the map update, the VPP takes new parameters collected from DERs and redoes the workflow shown in Fig. 5. Notice that the third and fourth steps in Fig. 5 should be conducted in a parallel way so that the updated process can be finished online before the market closes.

B. Marginal Cost of Increasing Latency

The marginal increasing cost due to the increasing latency is computed based on the aggregation cost minimization problem. It answers how much the cost will rise by ΔC , if DER k 's response latency increases marginally by $\Delta\tau_k$.

The response latency influences the aggregation cost via affecting the values of linear coefficients $\mathbf{a}$, $\mathbf{b}$ in Eqs. (17)-(18).

- The linear coefficients $\mathbf{a}$ represent the unit step responses of DERs at time $\tilde{t}_\phi^*$. Generally, $a_\phi < 0$ and $\mathbf{a}_l$, $\mathbf{a}_d > 0$, $\mathbf{a}_l$, $\mathbf{a}_d$ indicates DERs' capability to pull the system frequency at $\tilde{t}_\phi^*$ to the desired value, and thus are named *pulling ability*.
- The linear coefficients $\mathbf{b}$ represent the unit impulse responses of DERs at time $\tilde{t}_\phi^*$. Generally, $b_\phi > 0$ and $\mathbf{b}_l$, $\mathbf{b}_d < 0$. $\mathbf{b}_d$ represents DERs' capability to stop the system frequency after $\tilde{t}_\phi^*$ from declining, and thus are named *stopping ability*.

Take the effect of the increasing latency on DER k 's pulling ability for an example. When τ_k increases by $\Delta\tau_k > 0$, the linear coefficients a_k will change accordingly by Δa_k . Because $\partial a_k / \partial \tau_k < 0$, the pulling ability of DER k degrades $\Delta a_k < 0$ as the latency increases. In that case, the marginal change in Δa_k will possibly make the frequency nadir requirement constraint $|\Delta\omega(\tilde{t}_\phi^*) - \Delta\tilde{\omega}(\tilde{t}_\phi^*)| \leq \varepsilon_1$ violated. To make the new solution in the marginally perturbed problem feasible, the VPP has to increase the response power of other DERs by δP to keep the new solution feasible for the constraint $|\Delta\omega(\tilde{t}_\phi^*) - \Delta\tilde{\omega}(\tilde{t}_\phi^*)| \leq \varepsilon_1$, and it will consequently increase the aggregation cost by ΔC .

As explained above, the key to obtaining the marginal cost of latency is determining the marginal changes in the DER reserve portfolio δP to maintain the feasibility and optimality. Then the increasing marginal cost can be obtained via the marginal cost functions. Naturally, one could alter $\Delta\tau_k$ marginally, and then re-solve the optimization problem to get the new optimal solution $\mathbf{P}^*$. The cost difference between the old solution $\mathbf{P}^*$ and the new solution $\mathbf{P}^*$ is the ΔC , and the marginal cost of latency then can be given by $\Delta C / \Delta\tau_k$. This process will be repeated for every DER k . Nevertheless, this direct method is quite time-consuming. It requires the VPP to re-solve the optimization problem K times.

To accelerate the evaluation process, this paper proposes to use the sensitivity-based method. The sensitivity $\nabla_{\boldsymbol{\tau}} \mathbf{P}$ of the DER portfolio $\mathbf{P}$ with respect to the DERs' response latency $\boldsymbol{\tau}$ is directly computed without requiring re-solving the aggregation cost minimization problem. For the mixed-integer optimization problem, it is reasonable to assume that the solution of binary variables remains unchanged and stable within small perturbed latency [30]. After fixing the values of the binary variables $\mathbf{x}^*$, the aggregation cost minimization problem is expressed in a more compact form as follows:

$$\min_{\mathbf{P}_d} C \quad \text{s.t. } \mathbf{h}(\mathbf{P}_d, \boldsymbol{\tau}) \leq 0 : \boldsymbol{\gamma} \quad (19)$$

where $\boldsymbol{\gamma}$ represents the Lagrangian multipliers and $\mathbf{h}$ stands for the constraints. The decision variables in Eq. (19) are all

TABLE III
BASIC SETTING OF THE PARAMETERS

Parameter	Value
Governor time constant T_g/s	0.218
Steam chest time constant T_c/s	0.330
Reheat time constant T_r/s	9.722
High-pressure turbine fraction F_h/pu	0.265
Inverter time constant T_d/s	0.1
Inertia constant H/s	15.0
Governor speed regulation R	0.028
Load damping factor D	1.2

continuous, and its KKT condition can be expressed as:

$$\nabla_{P_d} \mathcal{L}(\mathbf{P}_d^*, \boldsymbol{\gamma}) = \nabla_{P_d} C + \text{diag}(\boldsymbol{\gamma}) \nabla_{P_d} \mathbf{h} = 0 \quad (20)$$

$$\text{diag}(\boldsymbol{\gamma}) \mathbf{h} = 0 \quad (21)$$

where Eq. (20) represents the stationary condition and Eq. (21) represents the complementary slackness requirement.

When latency changes marginally, the KKT condition Eqs. (20)-(21) still holds approximately [31]. So, the marginal changes $\nabla_{\boldsymbol{\tau}} \mathbf{P}_d$ can be obtained by computing the derivative concerning latency $\boldsymbol{\tau}$ of the KKT condition Eqs. (20)-(21) as:

$$\begin{bmatrix} \nabla_{P_d}^2 \mathcal{L} & \nabla_{P_d} \mathbf{h} \\ \Gamma (\nabla_{P_d} \mathbf{h})^T & \mathbf{H} \end{bmatrix} \begin{bmatrix} (\nabla_{\boldsymbol{\tau}} \mathbf{P}_d) \\ (\nabla_{\boldsymbol{\tau}} \boldsymbol{\gamma}) \end{bmatrix} = \begin{bmatrix} \nabla_{P_d \boldsymbol{\tau}}^2 \mathcal{L} \\ \Gamma (\nabla_{\boldsymbol{\tau}} \mathbf{h}) \end{bmatrix} \quad (22)$$

where $\Gamma = \text{diag}(\boldsymbol{\gamma})$ and $\mathbf{H} = \text{diag}(\mathbf{h})$. The first row is the derivative of Eq. (20) concerning the latency $\boldsymbol{\tau}$, and the second row is the derivative of Eq. (21) concerning the latency $\boldsymbol{\tau}$.

After obtaining $\nabla_{\boldsymbol{\tau}} \mathbf{P}_d$ via Eq. (22), the cost sensitivity $\nabla_{\boldsymbol{\tau}} C$ can be directly computed by taking $\nabla_{\boldsymbol{\tau}} \mathbf{P}_d$ into the marginal cost function of storage-type DERs as:

$$\nabla_{\boldsymbol{\tau}} C = (2\text{diag}(\boldsymbol{\rho}_d) \mathbf{P}_d^* + \boldsymbol{q}_d)^T \cdot (\nabla_{\boldsymbol{\tau}} \mathbf{P}_d) \quad (23)$$

By now, the marginal cost of latency can be computed via $\Delta C = \Delta \tau_k \times \nabla_{\tau_k} C$. Eqs. (22)-(23) are all in the matrix form. The most complicated part is to solve $\nabla_{\boldsymbol{\tau}} \mathbf{P}_d$ in Eq. (22). Nevertheless, the left-hand matrix in Eq. (22) is sparse and the solutions can be efficiently obtained.

V. CASE STUDIES

This section implements numerical experiments to verify the effectiveness of the proposed method. All the experiments are implemented using MATLAB software on a workstation with an Intel Core i9-10900K @ 3.70GHz CPU and 128.0 GB of RAM. The aggregation cost optimization problem is solved using Gurobi software.

The governor turbine & inverter dynamics $M_s(s)$ is set to be the sum of transfer functions of a reheat-type generator and a renewable generation:

$$M_s(s) = \frac{1}{1 + T_g s} \frac{1}{1 + T_c s} \frac{1 + F_h T_r s}{1 + T_r s} + \frac{1}{1 + T_d s} \frac{1}{5R^{-1}} \quad (24)$$

The model is an equivalent single-machine frequency model of all the thermal generators and renewable generations in the system, whose parameters can be obtained from [19]. Related parameters are listed in Table III. There are 5 poles in Eq. (1) and the poles are all simple complex poles with no repeat- ing. The value of p_i are: $p_1 = -9.9932$, $p_2 = -5.0210$, $p_3 =$

-2.2816 , $p_4(p_5) = -0.2366 + 0.3168j(-0.3168j)$. The values of A_i are: $A_1 = -0.0011$, $A_2 = 0.1098$, $A_3 = -0.5354$, $A_4(A_5) = -1.0467 + 0.4001j(-0.4001j)$. The rated base in the system is set to 1000 MW. For brevity, the power mentioned below are all in the unit of per unit base.

For DERs, 6000 storage-type DERs and 6000 load-type DERs are generated. The time constants of storage-type DERs are all set to $T_{d_k} = 0.1$ s. The maximum response power is randomly generated between $P_{\max} \in [10^{-6}, 10^{-5}]$ pu ($P_{\max} \in [1, 10]$ kW). DERs' response latency is randomly generated between $\tau_k \in [0, \tau_{\max}]$, where $\tau_{\max}$ is the maximum estimated latency. By default, $\tau_{\max} = 3000$ ms. The cost function of storage-type DERs is set as $q_{d_k} \in [0.025, 0.05] \times 10^6$ \$/pu ($q_{d_k} \in [0.025, 0.05]$ \$/kW) and $\rho_{d_k} = 2.5 \times 10^{10}$ \$/(pu)² ($\rho_{d_k} = 0.025$ \$/(kW)²). The cost function of load-type DERs is set as $q_{l_k} \in [0.05, 0.1] \times 10^6$ \$/pu ($q_{l_k} \in [0.05, 0.1]$ \$/kW).

The initial power imbalance P_ϕ is evenly distributed between $[0.08, 0.15]$ pu ([80, 150] MW) with an interval of 10^{-3} pu. Under this setting, the initial imbalance is around the power output of a thermal generating unit, and it will trigger contingency frequency control.

A. The Equivalent Aggregation Model

Different equivalent aggregation models in Table I are compared in the following cases:

- *Average Case*: Under this case, the *direct sum/average* method in Table I is used. The aggregated response for storage-types DERs and load-type DERs are computed separately. The aggregated power is the sum of all DERs. The aggregated time constant and the aggregated response delay are the weighted average of DERs according to the size of the reserve:

$$\tau_{\text{agg}} = \sum_k \frac{P_k}{\sum_{k'} P_{k'}} \cdot \tau_k, \quad T_{\text{agg}} = \sum_k \frac{P_k}{\sum_{k'} P_{k'}} \cdot T_k$$

The final aggregated response model is $\Delta P_0(s) = \frac{0.0332}{0.1s+1} e^{-1.5049s} + \frac{0.0328}{s} e^{-1.4929s}$, which contains 5 parameters.

- *Fitting case*: Under this case, a combination of the *fitting curve* method and the *optimization-based* method in Table I is used. The *curve fitting+sum* method cannot be used alone because the method requires a total 2K-3K parameters to define the aggregation model. The *optimization-based* method in [18] cannot be used solely because the model cannot account for response latency. Thereby, by combining the advantages of those two methods, we directly fit DERs' aggregated response power from $t = 0$ to $t = \infty$ to a first-order model. The combination method avoids high parameter complexity of the the *curve fitting+sum* method, and approximately minimizes the frequency errors from $t = 0$ to $t = \infty$ just as [18] did. The final aggregated response model is $\Delta P_0(s) = \frac{0.0660}{1.832s+1}$, which contains 2 parameters.
- *Equivalence case*: Under this case, the equivalent model in Def 2 is used. The penalized factor is set as $\lambda = 0.05$. In the gradient descent process, the initial values are set as $P_0 = 0.1$ pu and $T_0 = 5$ s; the descent step size is

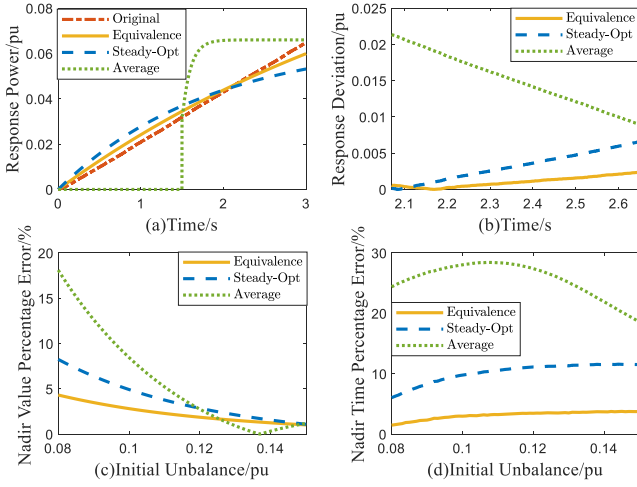


Fig. 6. (a) The original response power curve and the equivalent response power curve during 0–4 s; (b) The response deviations ($|\Delta P_0 - \sum_k \Delta P_k(s)|$) during possible nadir time; (c) The percentage error of nadir value $|\Delta \tilde{\omega}(\tilde{t}_\phi^*) - \Delta \omega(t_\phi^*)|/|\Delta \omega(t_\phi^*)|$; (d) The percentage error of nadir time $|\tilde{t}_\phi^* - t_\phi^*|/|t_\phi^*|$.

set as 100; the stop criterion is set as the norm of the gradient $\leq 10^{-9}$. The final aggregated response model is $\Delta P_0(s) = \frac{0.1309}{4.900s+1}$, which contains 2 parameters.

The comparison results are shown in Fig. 6. In Fig. 6(a), the original response power curve and the equivalent aggregation response curves obtained under three cases are exhibited. In Fig. 6(b), the response deviations between the equivalent aggregation response and the original response ($|\Delta P_0 - \sum_k \Delta P_k(s)|$) during possible nadir time t_ϕ^* are exhibited. The proposed equivalent model achieves the smallest differences with the original response among the three cases.

The percentage errors of nadir value ($|\Delta \tilde{\omega}(\tilde{t}_\phi^*) - \Delta \omega(t_\phi^*)|/|\Delta \omega(t_\phi^*)|$) and nadir time ($|\tilde{t}_\phi^* - t_\phi^*|/|t_\phi^*|$) under different initial imbalances $P_\phi \in [0.08, 0.15]$ pu are depicted in Fig. 6(c)-(d). The proposed equivalent aggregation model offers the closest approximation of the frequency nadir value and nadir time to the original response power. Although the average case achieves smaller nadir value differences under some values of P_ϕ , the nadir time is too far away from the true nadir time. In summary, Fig. 6 proves that the proposed equivalent aggregation model restores the DER aggregation's frequency support capability.

Next, the effect of response latency on the equivalent aggregation response is studied. The maximum latency $\tau_{\max}$ is scaled between $[1, 6]$ s. For example, when $\tau_{\max} = 6$ s, the latency of all DERs is scaled twice of that when $\tau_{\max} = 3$ s. When $\tau_{\max} = 1$ s, the latency of all DERs are scaled 1/3 of that when $\tau_{\max} = 3$ s. The solutions are shown in Fig. 7. Instead of exhibiting the direct solution value of P_0 , Fig. 7(b) shows the equivalent response power at $t = 2.25$ s, which is within the possible nadir time and reflects the VPP's actual contributions to the system frequency support. Because the value of P_0 corresponds to the steady response power at $t \rightarrow \infty$ (according to the final value theorem of the Laplace transform) and the equivalent aggregation model only captures the transient performance of DERs around the nadir time, comparisons among values of P_0 will not be helpful.

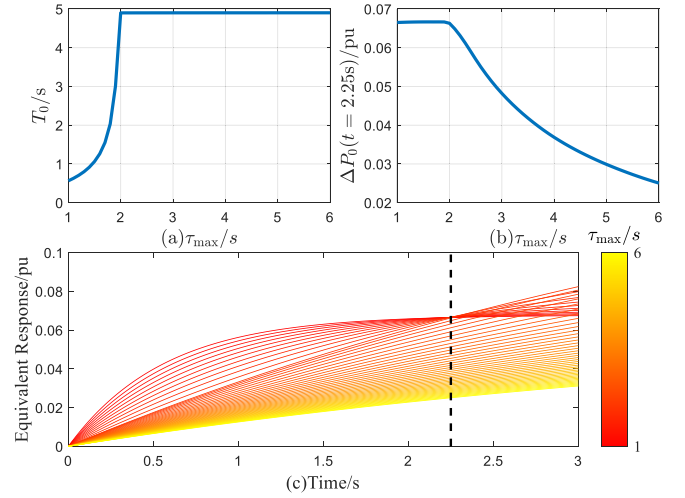


Fig. 7. (a) The solution T_0 . (b) The effective power response power $\Delta P_0(t = 2.25s)$. (c) The equivalent response curves $\Delta P_0(t)$.

When the response latency increases, there are two phases of changes. In the first phase (when $\tau_{\max} \in [1, 2]$ s), the equivalent time constant T_0 rises while the effective reserve power $\Delta P_0(t = 2.25s)$ remains unchanged. It indicates that the VPP can provide similar sizes of response power before the nadir time as the latency grows, but the response speed is getting slower. In the second phase (when $\tau_{\max} \in (2, 6]$ s), the equivalent time constant T_0 remains unchanged while the effective reserve power $\Delta P_0(t = 2.25s)$ decreases. This is because when $\tau_{\max} > 2$ s, the VPP cannot activate the same size of response power before the nadir time as $\tau_{\max} < 2$ s. Under this circumstance, the proposed equivalent aggregation model will lower the value of P_0 but rather than T_0 . This phenomenon can be further verified in Fig. 7(c), which clearly shows how the response curve varies when $\tau_{\max}$ varies. In summary, the equivalent aggregation model successfully captures the changes of the size and the speed of the aggregated response.

B. Performance-to-Cost Map

According to the work flow in Fig. 5, the highest-performance boundary point in the first step is already computed in the last subsection, $(P_{0(\max)}, T_{0(\min)}) = (0.1309 \text{ pu}, 4.9000 \text{ s})$. The lowest-performance boundary point in the second step is selected as $(P_{0(\min)}, T_{0(\max)}) = (0.2P_{0(\max)}, 3T_{0(\min)})$, and the parameter space granularity is selected as $\delta P_0 = 0.02P_{0(\max)}$, $\delta T_0 = 0.05T_{0(\min)}$. Under this setting, the map grid is composed of $41 \times 41 = 1681$ pairs of P_0, T_0 .

The third and the fourth steps in Fig. 5 are conducted parallelly via parallel workers in MATLAB software. There are in total of 20 parallel worker running the third and the fourth steps simultaneously. A single aggregation cost minimization problem program can be solved in 5 seconds with a feasibility tolerance of 10^{-9} . According to the timing results, the first step to the fourth step cost 63 seconds, 0.01 seconds, 33 seconds, and 615 seconds, respectively. The total map update time is 711 seconds, which is less than 15 minutes. It proves that the map can be updated in one market period.

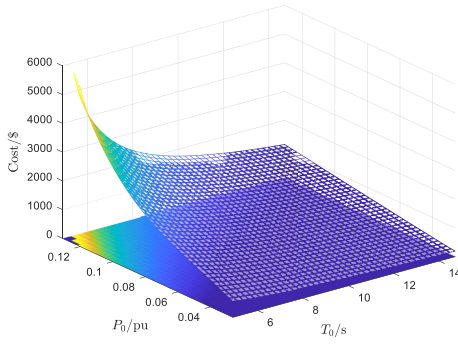


Fig. 8. The performance-to-cost map when $\tau_{\max} = 3$ s.

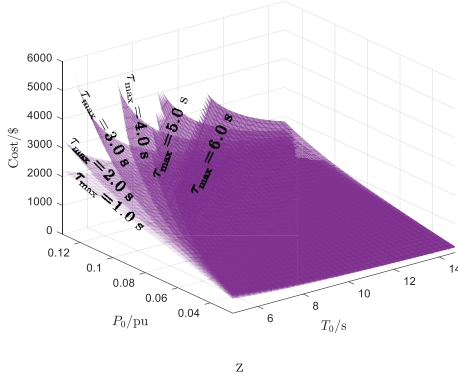


Fig. 9. The performance-to-cost map when $\tau_{\max} = 1, 2, 3, 4, 5, 6$ s.

The final result of the aggregation cost map is shown in Fig. 8, where the maximum latency is set as the default value $\tau_{\max} = 3$ s. In Fig. 8, when P_0 rises and T_0 lowers, the requirements of the aggregation's transient performance are higher, and it also incurs a higher aggregation cost for the VPP to deliver the required contingency reserve. As the value P_0 and T_0 reaches the limit $P_0 \rightarrow P_{0(\max)}$ and $T_0 \rightarrow T_{0(\min)}$, the aggregation cost rises more aggressively.

The changes of the cost map under different maximum latency $\tau_{\max} \in \{1, 2, 3, 4, 5, 6\}$ s are shown in Fig. 9, which reflect the global effects of latency on the aggregation cost. Firstly, a higher latency restricts the performance of the contingency reserve that the VPP can provide. The range of P_0 and T_0 shrinks as latency rises. For instance, when $\tau_{\max} = 4, 5, 6$ s, the solution map cannot reach the pair of (0.1309 pu, 4.9000 s). The Gurobi solver directly returns 'no solution' then. It proves that the equivalence Def 2 indeed finds the highest-performance reserve that the VPP can provide. On the other hand, as the latency rises, the cost to provide a certain pair of P_0 and T_0 increases. At the highest-performance reserve (0.1309 pu, 4.9000 s), the VPP's aggregation cost rises around 4000 \$ when latency is $\tau_{\max} = 3.0$ s than that when the latency is $\tau_{\max} = 1.0$ s. The results above highlight the importance of low-latency communication infrastructure for the VPP to provide contingency reserves.

C. Marginal Cost of Latency $\partial C / \partial \tau_k$

To prove that the marginal cost of latency computed through the sensitivity-based method is correct, this paper conducts the following experiments:

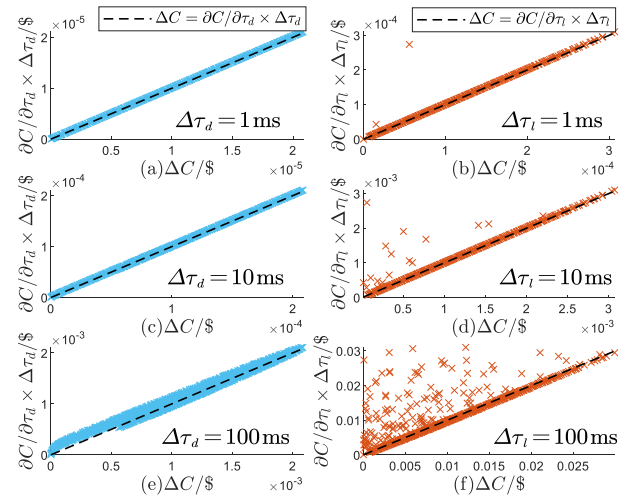


Fig. 10. The increasing cost obtained by the direct solution method and the marginal approximation method under different latency increase margins. (a): $\Delta \tau_d = 1$ ms for storage-type DERs. (b): $\Delta \tau_l = 1$ ms for load-type DERs. (c): $\Delta \tau_d = 10$ ms for storage-type DERs. (d): $\Delta \tau_l = 10$ ms for load-type DERs. (e): $\Delta \tau_d = 100$ ms for storage-type DERs. (f): $\Delta \tau_l = 100$ ms for load-type DERs.

- Direct solution method: increase DER k 's latency by $\Delta \tau_k$ each time and re-solve the cost minimization problem. The cost difference between the new solution and the original solution is ΔC .
- Marginal approximation method: compute the marginal cost $\partial C / \partial \tau_k$ based on the sensitivity-based method and computes the cost increase as $\partial C / \partial \tau_k \times \Delta \tau_k$.

The direct solution method offers an accurate cost increase, while the marginal approximation method offers an approximate cost increase. Three latency increase $\Delta \tau_k$ settings are computed: $\Delta \tau_k = 1$ ms, $\Delta \tau_k = 10$ ms, $\Delta \tau_k = 100$ ms. Direct solution method requires to solve the optimization problem for all of DERs for 12000 times. Even with parallel workers, it takes more than 1 hour to get the final result ΔC for all DERs. And if $\Delta \tau_k$ changes, the result must be re-computed. On the contrary, the marginal approximation method takes less than 5 seconds to get the result $\partial C / \partial \tau_k \times \Delta \tau_k$.

The comparison result is shown in Fig. 10. When $\Delta \tau_k = 1$ ms, the points are located on the line $\Delta C = \partial C / \partial \tau_k \times \Delta \tau_k$, which means the two results are nearly equal. It proves that the marginal cost of latency computed through sensitivity analysis is accurate enough. When $\Delta \tau_k$ increases, there are some points deviating from the line $\Delta C = \partial C / \partial \tau_k \times \Delta \tau_k$ because the integer solution changes after increasing the latency. The values of mean absolute percentage errors (MPAE) $|\partial C / \partial \tau_k \times \Delta \tau_k - \Delta C| / \Delta C$ under the six scenarios are computed to show the effectiveness of the sensitivity-based method. From top to bottom, from left to the right, the values of MPAE are: (a) 1.3703%, (b) 1.4504%, (c) 4.3663%, (d) 5.9461%, (e) 5.7276%, (f) 8.0119%. It shows that although the result of the sensitivity-based method deviates from the accurate cost changes, it is still a qualified approximation when latency increases.

Further analysis is conducted to explain how latency influences the aggregation cost, as shown in Fig. 11. In Fig. 11(a),

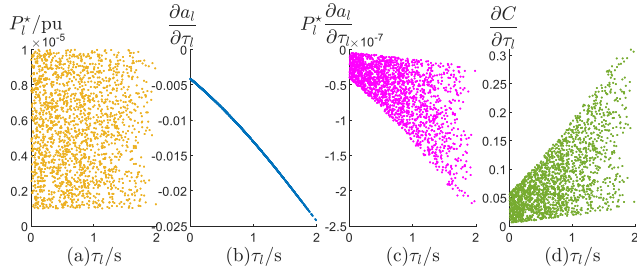


Fig. 11. Marginal cost of latency analysis for load-type DERs.

it is shown that the value of the solution P_l^* is unrelated to latency τ_l . This is because load-type DERs can only be switched between the on and off states, and the value of P_l^* can not be adjusted freely. Nevertheless, it is shown that the number of selected load-type DERs decreases as the latency increases, because the density of the points gets thicker. In Fig. 11(b), it is shown that the values of $\partial a_l / \partial \tau_l$ are all negative and its absolute value grows as the latency τ_l rises. Combining the results of Fig. 11(a) and (b), the products of $\partial a_l / \partial \tau_l$ and P_l^* are negative and also the absolute value tends to increase as the latency τ_l rises, shown as Fig. 11(c). The changes of $\Delta\omega(\tilde{\tau}_\phi^*)$ in Eq. (17) when latency τ_l changes can be approximated by $P_l^* \partial a_l / \partial \tau_l \times \Delta\tau_l$. Thereby, a larger $P_l^* \partial a_l / \partial \tau_l$ results in a larger change in the constraint value and requires for more power to keep the constraint feasible. In other words, they incur more power to fix the performance degradation, and thus the load-type DERs with higher latency incur more marginal cost when latency increases, as shown in Fig. 11(d).

VI. CONCLUSION AND FUTURE WORKS

The paper investigates how to involve DERs in contingency frequency support through VPPs. A technique of finding the equivalent aggregation model for DERs is offered; the performance-to-cost map is built via the cost minimization optimization problem; the marginal cost of latency is provided to evaluate DERs' response abilities. The numerical results demonstrate the effectiveness of the equivalent aggregation model and the method to compute the marginal cost of latency. The results identify the significance to investigate economic effects of cyber systems (including measurement systems and communication systems) on power system operations.

This paper does not consider the effects of networks on the response performance, which is an important factor to be considered due to the negligible resistive loss in the distribution grid. Besides, further studies should also be conducted to investigate how to coordinate DERs for more types of services while considering the effects of cyber systems on them. The VPP can actively adjust the operation of the cyber system and minimize its adverse effects on these services. Therefore, it is necessary to investigate how to perform optimizations on the operation of cyber systems to cater to the VPP's needs.

APPENDIX A

The gradients in the gradient descent algorithm for solving Eq. (11) are:

$$\frac{\partial \Delta\tilde{\omega}}{\partial P_0} = \sum_i \left[\frac{A_i}{p_i} (e^{p_i t_\phi^*} - 1) - \frac{A_i}{p_i + T_0^{-1}} (e^{p_i t_\phi^*} - e^{-T_0^{-1} t_\phi^*}) \right]$$

$$\frac{\partial \Delta\tilde{\omega}}{\partial T_0} = P_0 \sum_i \frac{A_i}{T_0^2} \left[\frac{t_\phi^* e^{-T_0^{-1} t_\phi^*}}{(p_i + T_0^{-1})} - \frac{e^{p_i t_\phi^*} - e^{-T_0^{-1} t_\phi^*}}{(p_i + T_0^{-1})^2} \right]$$

$$\frac{\partial \Delta\dot{\omega}}{\partial P_0} = \sum_i \frac{A_i}{p_i T_0 + 1} [e^{p_i t_\phi^*} - e^{-T_0^{-1} t_\phi^*}]$$

$$\frac{\partial \Delta\dot{\omega}}{\partial T_0} = P_0 \cdot$$

$$\sum_i -\frac{A_i}{(p_i T_0 + 1)^2} \left[p_i (e^{p_i t_\phi^*} - e^{-T_0^{-1} t_\phi^*}) + \frac{p_i T_0 + 1}{T_0^2} t_\phi^* e^{-T_0^{-1} t_\phi^*} \right]$$

APPENDIX B

The Lagrangian function is:

$$\begin{aligned} \mathcal{L}(\mathbf{P}_d, \bar{\boldsymbol{\mu}}, \underline{\boldsymbol{\mu}}, \bar{\mathbf{v}}, \underline{\mathbf{v}}, \bar{\boldsymbol{\alpha}}, \underline{\boldsymbol{\alpha}}) = & \sum_{k \in d_k} c_k(P_k) + \\ & \sum_m \bar{\boldsymbol{\mu}}_m (\mathbf{a}_{d(m)}^T \mathbf{P}_d + \mathbf{a}_{l(m)}^T \mathbf{P}_l^* + a_{\phi(m)} - \Delta\tilde{\omega}(\tilde{\tau}_{\phi(m)}^*) - \varepsilon_1) - \\ & \sum_m \underline{\boldsymbol{\mu}}_m (\mathbf{a}_{d(m)}^T \mathbf{P}_d + \mathbf{a}_{l(m)}^T \mathbf{P}_l^* + a_{\phi(m)} - \Delta\tilde{\omega}(\tilde{\tau}_{\phi(m)}^*) + \varepsilon_1) + \\ & \sum_m \bar{\mathbf{v}}_m (\mathbf{b}_{l(m)}^T \mathbf{P}_l^* + \mathbf{b}_{d(m)}^T \mathbf{P}_d + b_{\phi(m)} - \varepsilon_2) - \\ & \sum_m \underline{\mathbf{v}}_m (\mathbf{b}_{l(m)}^T \mathbf{P}_l^* + \mathbf{b}_{d(m)}^T \mathbf{P}_d + b_{\phi(m)} + \varepsilon_2) + \\ & + \text{diag}(\bar{\boldsymbol{\alpha}})(\mathbf{P}_d - \mathbf{P}_{\max}) - \text{diag}(\underline{\boldsymbol{\alpha}})\mathbf{P}_d \end{aligned}$$

where m stands for constraints with m th initial imbalance ϕ_m .

APPENDIX C

The expression of the left-hand matrix in Eq. (22) is:

$$\begin{aligned} \nabla_{\mathbf{P}_d}^2 \mathcal{L} &= \text{diag}(\partial^2 c_{d_k} / \partial P_{d_k}^2) = \text{diag}(2 * \boldsymbol{\rho}_d) \\ \nabla_{\mathbf{P}_d} \mathbf{h} &= [\mathbf{A} \quad -\mathbf{A} \mathbf{B} \quad -\mathbf{B} \mathbf{E}_{d_K} \quad -\mathbf{E}_{d_K}] \\ \Gamma &= \text{diag}([\bar{\boldsymbol{\mu}}, \underline{\boldsymbol{\mu}}, \bar{\mathbf{v}}, \underline{\mathbf{v}}, \bar{\boldsymbol{\alpha}}, \underline{\boldsymbol{\alpha}}]) \\ \mathbf{H} &= \text{diag}([h_1, h_2, h_3, h_4, h_5, h_6]) \\ h_{1(m)} &= \mathbf{a}_{d(m)}^T \mathbf{P}_d + \mathbf{a}_{l(m)}^T \mathbf{P}_l^* + a_{\phi(m)} - \Delta\tilde{\omega}(\tilde{\tau}_{\phi(m)}^*) - \varepsilon_1 \\ h_{2(m)} &= -(\mathbf{a}_{d(m)}^T \mathbf{P}_d + \mathbf{a}_{l(m)}^T \mathbf{P}_l^* + a_{\phi(m)} - \Delta\tilde{\omega}(\tilde{\tau}_{\phi(m)}^*) + \varepsilon_1) \\ h_{3(m)} &= \mathbf{b}_{l(m)}^T \mathbf{P}_l^* + \mathbf{b}_{d(m)}^T \mathbf{P}_d + b_{\phi(m)} - \varepsilon_2 \\ h_{4(m)} &= -(\mathbf{b}_{l(m)}^T \mathbf{P}_l^* + \mathbf{b}_{d(m)}^T \mathbf{P}_d + b_{\phi(m)} + \varepsilon_2) \\ h_5 &= \mathbf{P}_d - \mathbf{P}_{\max}, h_6 = -\mathbf{P}_d \end{aligned}$$

where $\mathbf{A}$ is the matrix with the element $a_{k,d(m)}$ at the m th row and k th column; $\mathbf{B}$ is the matrix with the element $b_{k,d(m)}$ at the m th row and k th column. $\mathbf{1}_K$ stands for a column vector composed of K elements all equal to 1; $\mathbf{E}_{d_K}$ stands for the identity matrix with size d_K .

For the expression of right-hand matrix in Eq. (22), if $\boldsymbol{\tau}$ is $\boldsymbol{\tau}_d$, $\nabla_{\boldsymbol{\tau}_d}^2 \mathcal{L}$ is a diagonal matrix in size of K_d and the k th

diagonal elements are:

$$\sum_m \left(\bar{\mu}_m - \underline{\mu}_m \right) \frac{\partial a_{d(m),k}}{\partial \tau_k} + \left(\bar{\nu}_m - \underline{\nu}_m \right) \frac{\partial b_{d(m),k}}{\partial \tau_k}$$

If τ is τ_l , $\nabla_{\tau}^2 \mathcal{L}$ is a zeros matrix in size of K_l .

The other sub-matrix are as follows:

$$\begin{aligned} \nabla_{\tau} \mathbf{h} &= \left[(\nabla_{\tau} \mathbf{h}_1)^T (\nabla_{\tau} \mathbf{h}_2)^T (\nabla_{\tau} \mathbf{h}_3)^T (\nabla_{\tau} \mathbf{h}_4)^T \mathbf{0} \mathbf{0} \right]^T \\ \nabla_{\tau} h_{1(m),k} &= P_k \frac{\partial a_{d(m),k}}{\partial \tau_k}, \nabla_{\tau} \mathbf{h}_2 = -\nabla_{\tau} \mathbf{h}_1 \\ \nabla_{\tau} h_{3(m),k} &= P_k \frac{\partial b_{d(m),k}}{\partial \tau_k}, \nabla_{\tau} \mathbf{h}_4 = -\nabla_{\tau} \mathbf{h}_3 \end{aligned}$$

where $\nabla_{\tau} h_{1(m),k}$ is the element at the k th row and the m th column.

REFERENCES

- [1] D. Fernández-Munoz, J. I. Pérez-Díaz, I. Guisández, M. Chazarra, and Á. Fernández-Espina, "Fast frequency control ancillary services: An international review," *Renew. Sustain. Energy Rev.*, vol. 120, Mar. 2020, Art. no. 109662.
- [2] H. Bevrani, *Robust Power System Frequency Control* (Power Electronics and Power Systems). New York, NY, USA: Springer, 2008.
- [3] B. Mohandes, M. S. El Moursi, N. Hatziargyriou, and S. El Khatib, "A review of power system flexibility with high penetration of renewables," *IEEE Trans. Power Syst.*, vol. 34, no. 4, pp. 3140–3155, Jul. 2019.
- [4] H. Hui, Y. Ding, Q. Shi, F. Li, Y. Song, and J. Yan, "5G network-based Internet of Things for demand response in smart grid: A survey on application potential," *Appl. Energy*, vol. 257, Jan. 2020, Art. no. 113972.
- [5] J. Schipper, A. Wood, and C. Edwards, "Optimizing instantaneous and ramping reserves with different response speeds for contingencies—Part I: Methodology," *IEEE Trans. Power Syst.*, vol. 35, no. 5, pp. 3953–3960, Sep. 2020.
- [6] H. Hui, Y. Ding, Y. Song, and S. Rahman, "Modeling and control of flexible loads for frequency regulation services considering compensation of communication latency and detection error," *Appl. Energy*, vol. 250, pp. 161–174, Sep. 2019.
- [7] Z. Yi, Y. Xu, W. Gu, and W. Wu, "A multi-time-scale economic scheduling strategy for virtual power plant based on deferrable loads aggregation and disaggregation," *IEEE Trans. Sustain. Energy*, vol. 11, no. 3, pp. 1332–1346, Jul. 2020.
- [8] S. Wang and W. Wu, "Aggregate flexibility of virtual power plants with temporal coupling constraints," *IEEE Trans. Smart Grid*, vol. 12, no. 6, pp. 5043–5051, Nov. 2021.
- [9] H. Zhao et al., "Active dynamic aggregation model for distributed integrated energy system as virtual power plant," *J. Modern Power Syst. Clean Energy*, vol. 8, no. 5, pp. 831–840, Sep. 2020.
- [10] I. Avramidis, F. Capitanescu, and G. Deconinck, "Grid-friendly smart sustainable buildings: Flexibility-to-cost mapping," *IEEE Trans. Sustain. Energy*, vol. 13, no. 3, pp. 1857–1860, Jul. 2022.
- [11] R. Ghorani, M. Fotuhi-Firuzabad, and M. Moeini-Aghaie, "Main challenges of implementing penalty mechanisms in transactive electricity markets," *IEEE Trans. Power Syst.*, vol. 34, no. 5, pp. 3954–3956, Sep. 2019.
- [12] H. Früh, S. Müller, D. Contreras, K. Rudion, A. von Haken, and B. Surmann, "Coordinated vertical provision of flexibility from distribution systems," *IEEE Trans. Power Syst.*, early access, Mar. 24, 2022, doi: 10.1109/TPWRS.2022.3162041.
- [13] S. Riaz and P. Mancarella, "Modelling and characterisation of flexibility from distributed energy resources," *IEEE Trans. Power Syst.*, vol. 37, no. 1, pp. 38–50, Jan. 2022.
- [14] Z. Tan, H. Zhong, X. Wang, and H. Tang, "An efficient method for estimating capability curve of virtual power plant," *CSEE J. Power Energy Syst.*, vol. 8, no. 3, pp. 780–788, May 2022.
- [15] X. Chen and N. Li, "Leveraging two-stage adaptive robust optimization for power flexibility aggregation," *IEEE Trans. Smart Grid*, vol. 12, no. 5, pp. 3954–3965, Sep. 2021.
- [16] J. Chen, M. Liu, and F. Milano, "Aggregated model of virtual power plants for transient frequency and voltage stability analysis," *IEEE Trans. Power Syst.*, vol. 36, no. 5, pp. 4366–4375, Sep. 2021.
- [17] E. O. Kontis, Á. R. Del Nozal, J. M. Mauricio, and C. S. Demoulias, "Provision of primary frequency response as ancillary service from active distribution networks to the transmission system," *IEEE Trans. Smart Grid*, vol. 12, no. 6, pp. 4971–4982, Nov. 2021.
- [18] S. S. Guggilam, C. Zhao, E. Dall'Anese, Y. C. Chen, and S. V. Dhople, "Optimizing DER participation in inertial and primary-frequency response," *IEEE Trans. Power Syst.*, vol. 33, no. 5, pp. 5194–5205, Sep. 2018.
- [19] Q. Shi, F. Li, and H. Cui, "Analytical method to aggregate multi-machine SFR model with applications in power system dynamic studies," *IEEE Trans. Power Syst.*, vol. 33, no. 6, pp. 6355–6367, Nov. 2018.
- [20] L. Liu, W. Li, Y. Ba, J. Shen, C. Jin, and K. Wen, "An analytical model for frequency nadir prediction following a major disturbance," *IEEE Trans. Power Syst.*, vol. 35, no. 4, pp. 2527–2536, Jul. 2020.
- [21] G. Zhang, J. McCalley, and Q. Wang, "An AGC dynamics-constrained economic dispatch model," *IEEE Trans. Power Syst.*, vol. 34, no. 5, pp. 3931–3940, Sep. 2019.
- [22] L. Badesa, F. Teng, and G. Strbac, "Optimal portfolio of distinct frequency response services in low-inertia systems," *IEEE Trans. Power Syst.*, vol. 35, no. 6, pp. 4459–4469, Nov. 2020.
- [23] C. Li, C. Duan, and Y. Cao, "An efficient method for computing exact delay-margins of large-scale power systems," *IEEE Trans. Power Syst.*, vol. 35, no. 6, pp. 4924–4927, Nov. 2020.
- [24] *Settlements Guide to Ancillary Services Payment and Recovery*, AEMO, Melbourne, VIC, Australia, Feb. 2020.
- [25] S. Baros, Y. C. Chen, and S. V. Dhople, "Examining the economic optimality of automatic generation control," *IEEE Trans. Power Syst.*, vol. 36, no. 5, pp. 4611–4620, Sep. 2021.
- [26] H. Li, Y. Qiao, Z. Lu, B. Zhang, and F. Teng, "Frequency-constrained stochastic planning towards a high renewable target considering frequency response support from wind power," *IEEE Trans. Power Syst.*, vol. 36, no. 5, pp. 4632–4644, Sep. 2021.
- [27] A. J. Wood, B. F. Wollenberg, and G. B. Sheblé, *Power Generation, Operation, and Control*. Hoboken, NJ, USA: Wiley, 2013.
- [28] D. L. H. Aik, "A general-order system frequency response model incorporating load shedding: Analytic modeling and applications," *IEEE Trans. Power Syst.*, vol. 21, no. 2, pp. 709–717, May 2006.
- [29] D. G. Luenberger and Y. Ye, *Linear and Nonlinear Programming*, vol. 2. New York, NY, USA: Springer, 1984.
- [30] M. Guzelsoy and T. K. Ralphs, "Duality for mixed-integer linear programs," *Int. J. Oper. Res.*, vol. 4, no. 3, pp. 118–137, 2007.
- [31] A. V. Fiacco, "Sensitivity analysis for nonlinear programming using penalty methods," *Math. Program.*, vol. 10, no. 1, pp. 287–311, 1976.



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